

# AI-Powered Harvest Yield Prediction for Non-Cash Crops (Big Onion) in Sri Lanka

Pre-harvest, district-level yield forecasting from open weather, satellite and soil data — with calibrated uncertainty.

## 1 Problem Statement

Sri Lanka consumes roughly **220,000 metric tons** of big onion each year. Domestic cultivation covers only part of that demand and the balance is imported at a direct foreign-exchange cost.

Unlike paddy, big onion has **no crop-cutting survey methodology**. Yield figures are compiled from the subjective assessments of agricultural officers, they carry no quantified error, and they are finalised only **after** the crop has been harvested and sold — while import volumes must be committed months in advance.



## 2 Project Objectives

- Data foundation** — integrate district yield, daily weather, satellite vegetation and soil data into one clean, reproducible dataset.
- Feature engineering** — derive agronomically meaningful predictors of thermal, water and canopy status.
- Comparative modelling** — train classical, deep, symbolic and physics-hybrid models under one identical protocol.
- Honest evaluation** — leakage-free cross-validation, paired significance testing and calibrated prediction intervals.
- Attribution and ablation** — quantify what each data source and each predictor actually contributes.
- Delivery** — a documented REST service and a decision-support dashboard for non-technical users.

## 3 Methodology

**Data.** Four open streams — DCS district yield, NASA POWER and CHIRPS weather, MODIS and Sentinel-2 vegetation and land-surface temperature via Google Earth Engine, and ISRIC SoilGrids — are reduced to one record per district, season and year.

**Features.** 22 predictors derived from 37,988 real daily weather records. Seven were removed as constants or exact linear transforms of others, and the yield lags were dropped as unfixable leaks under LOYO.

**Models.** Trained identically against two references — the train mean (the bar) and an oracle year-mean (not achievable, bounding what perfect year knowledge buys) — plus Random Forest, XGBoost, SVR and **PADR**, a 17-parameter agronomic model whose crop constants are estimated rather than assumed.

**Evaluation. Leave-One-Year-Out cross-validation:** an entire year is held out, hyperparameters are tuned inside the training partition only, and every fitted quantity is re-estimated per fold. A random split would leak information between districts of the same year and flatter the result. Models are compared with the paired Wilcoxon signed-rank test, and split-conformal intervals are attached to every forecast.

## 4 Key Modules & Components

| Module            | Responsibility                                                                               | Owner        |
|-------------------|----------------------------------------------------------------------------------------------|--------------|
| Data pipeline     | Source ingestion, key reconciliation, season windowing, cleaning, 32-predictor engineering   | Sharuja B.   |
| Prediction engine | Nine models, LOYO-CV harness, stacking, conformal calibration, SHAP attribution              | Arkam B.H.M. |
| REST service      | Validated prediction endpoint with interval, attribution and provenance; 8 endpoints         | Arkam B.H.M. |
| Dashboard         | Choropleth overview, prediction form with context prefill, explainability and advisory views | Shathurya P. |

## 5 Research Novelty

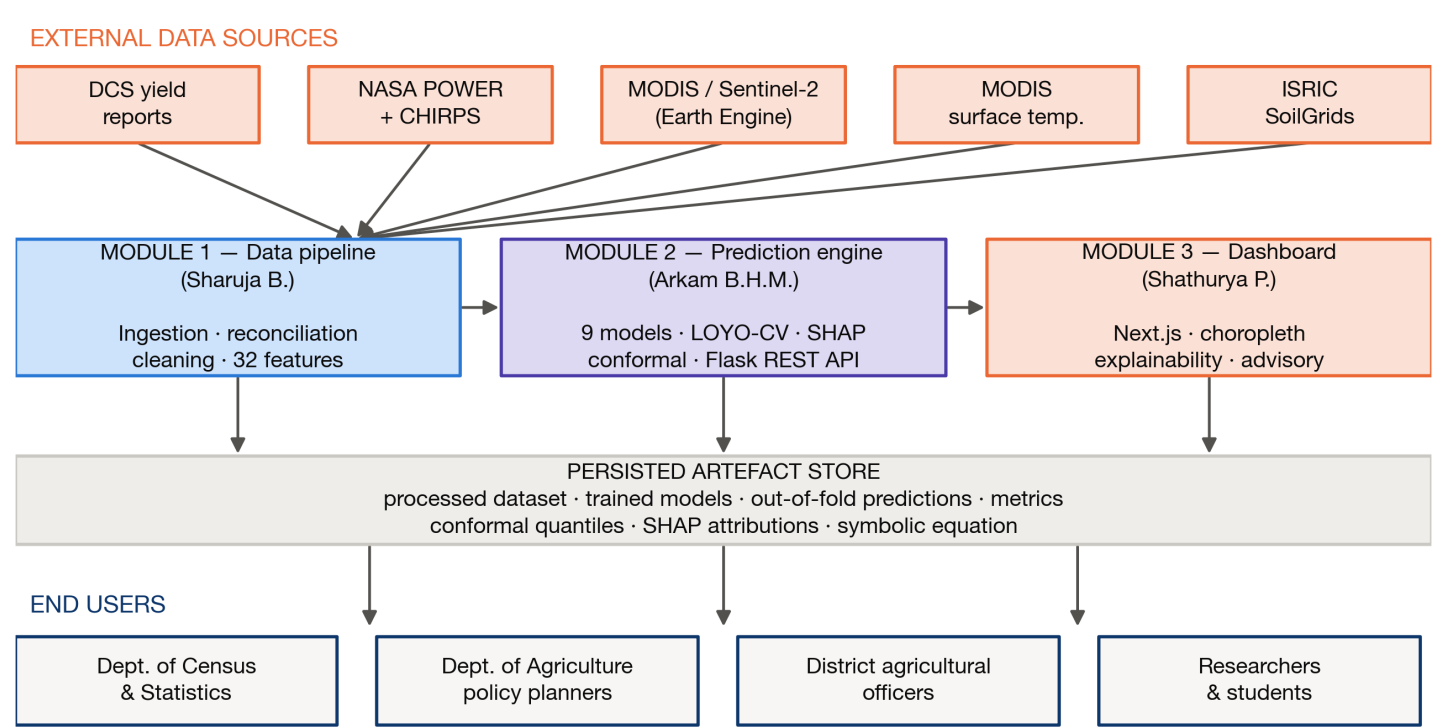
### 1 — The agronomic constants are ESTIMATED, not fixed

Existing hybrids freeze the crop model at published coefficients and fit a learner to its residual. PADR estimates  $K_y$ ,  $T_{\text{base}}$ ,  $T_{\text{opt}}$ ,  $T_{\text{crit}}$  and soil water capacity jointly, under agronomic box bounds, with an L2 penalty shrinking each toward its textbook value. +0.566  $R^2$ .

### 2 — Weather on a thermal clock, anchored on reported harvest

Onion is 0.002–0.89 % of any district, so satellite phenology fails: 25 of 28 district-years would not anchor. Instead the harvest date comes from the DCS monthly production record and planting is located by accumulating degree-days backwards. 28 of 28 anchored.

## 6 System Architecture



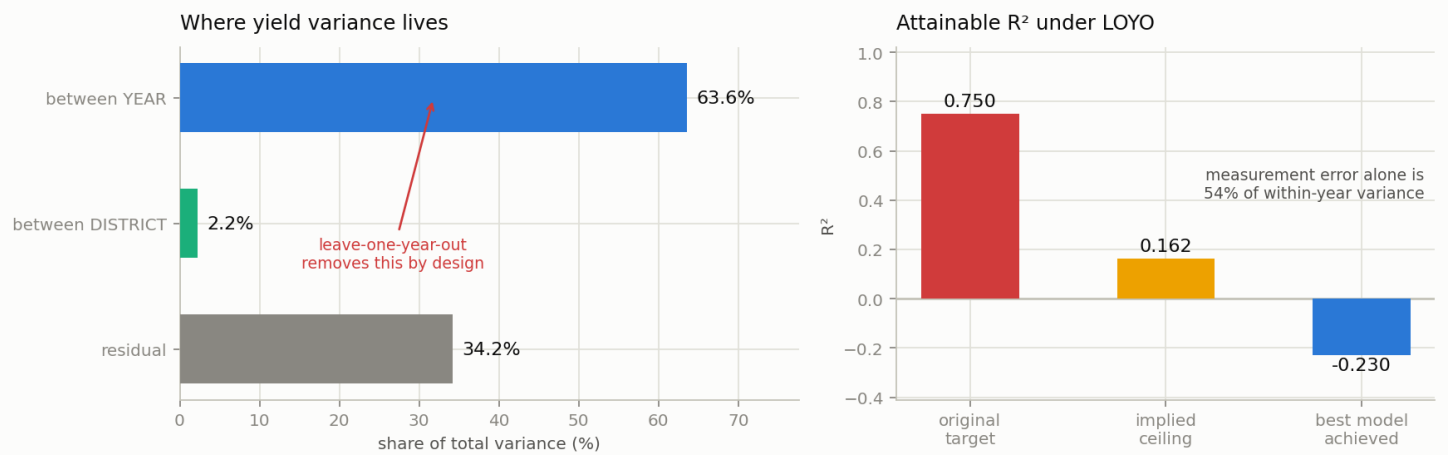
Four external data streams feed three build modules; the serving layer reads persisted artefacts and never trains.

## 7 Key Results

### HEADLINE — the $R^2 > 0.75$ target was unattainable, not merely unmet

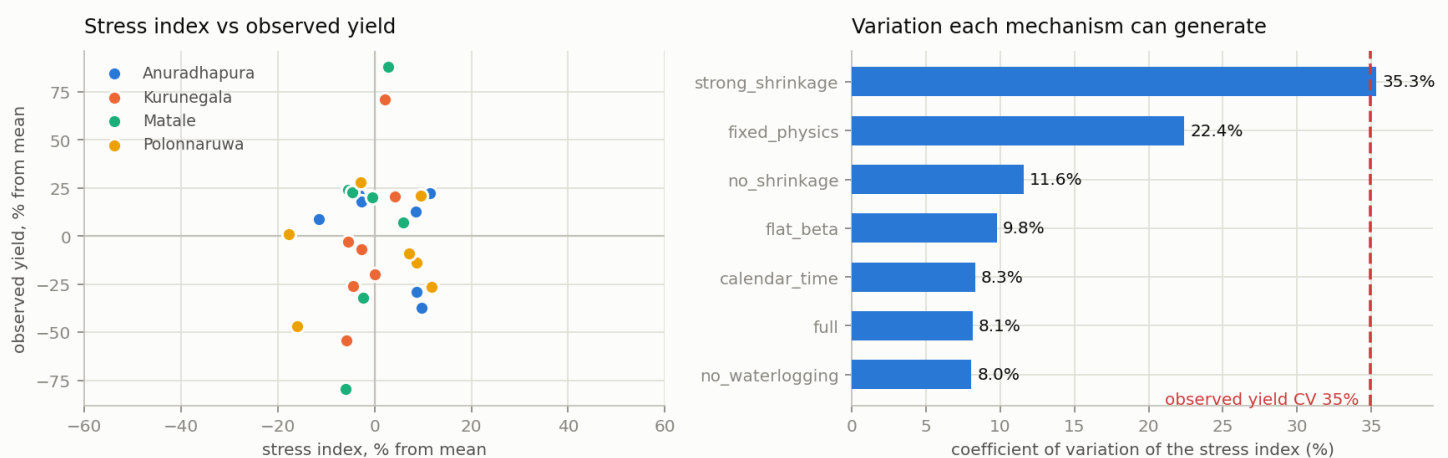
64 % of variance between years · removed by LOYO by construction · measurement error 54 % of the rest ·   
Leave-One-Year-Out over 28 corrected records (4 districts × 7 years, Yala). Every model scores below the train mean.

The  $R^2 > 0.75$  target was not difficult — it was unattainable



64 % of variance lies between years and LOYO removes it by design; measurement error takes 54 % of what remains within a year.

A model whose output varies 8% cannot explain a target that varies 35%



The stress index varies 8 % while observed yield varies 35 % — the agro-climatic channel cannot carry the signal.

## 8 Outcomes & Conclusion

- We audited our own dataset and it did not survive.** 40 % of the target was fabricated — 50 of 124 month-rows — and the fabrication reached the target, not just the covariates. Rebuilt from 87 genuine DCS records; the corrected target correlates with the old one at 0.68.
- The accuracy target was unattainable.** With 64 % of variance between years and measurement error at 54 % of the rest, no model of any architecture could exceed  $R^2$  0.162 under this protocol.
- Big onion here is not agro-climatically limited.** The stress index varies 8 % against 35 % in observed yield. Thermal stress is near-constant and water deficit almost never binds under tank irrigation. Removing the anomalous years makes the fit relatively worse, so the signal is absent.
- The instrument was sensitive enough to be believed.** On simulated panels PADR recovers a weather signal driving as little as 6 % of yield variance at  $n = 28$ . It recovered none from the real data, so the true agro-climatic signal is below 6 % of variance.
- FAO-33 overstates weather sensitivity for this crop.** Pinning the coefficients to published values gives the right variation (35 %) but the worst score of any arm ( $R^2 -1.54$ ). Estimating them locally removes spurious sensitivity — an argument for local calibration. Recovery testing confirms  $T_{\text{opt}}$  and  $W_{\text{max}}$  are identifiable at this sample size;  $K_y$  and  $T_{\text{crit}}$  are not, and are not claimed.

### Faculty of Information Technology

Department of Information Technology  
B.Sc. (Hons) in Information Technology  
University of Moratuwa, Sri Lanka

### GROUP MEMBERS

**214019K** Arkam B.H.M.  
**214192G** Sharuja B.  
**214193K** Shathurya P.

### PROJECT SUPERVISOR

**Dr. Firdhous M.F.M.**  
Faculty of Information Technology,  
University of Moratuwa

# AI-Powered Harvest Yield Prediction for Non-Cash Crops (Big Onion) in Sri Lanka

Pre-harvest, district-level yield forecasting from open weather, satellite and soil data — with calibrated uncertainty.

- **Next** — collect fertilizer, irrigation and price data. Our finding says precisely where the signal is not, and it is not in more weather.

Faculty of Information Technology

Department of Information Technology  
B.Sc. (Hons) in Information Technology  
University of Moratuwa, Sri Lanka

GROUP MEMBERS

|         |              |
|---------|--------------|
| 214019K | Arkam B.H.M. |
| 214192G | Sharuja B.   |
| 214193K | Shathurya P. |

PROJECT SUPERVISOR

**Dr. Firdhous M.F.M.**  
Faculty of Information Technology,  
University of Moratuwa